

Advanced (Habanero)

- Q1.** What is the solution to the system $x+y=2$ and $x-y=-2$?
(A) $(-2, 4)$
(B) $(2, 0)$
(C) $(0, -2)$
(D) $(1, 1)$
(E) No solution
- Q2.** Which of the following systems has no solution?
(A) $x+y=2$ and $x+y=3$
(B) $x-y=1$ and $x+y=2$
(C) $2x+2y=4$ and $x+y=2$
(D) $x+y=1$ and $x-y=-1$
(E) $y=2x+1$ and $y=2x-1$
- Q3.** The system $x+y=4$ and $x-y=0$ has a solution at
(A) $(2, 2)$
(B) $(0, 4)$
(C) $(4, 0)$
(D) $(1, 1)$
(E) $(3, 1)$
- Q4.** Which of the following is a true statement about the system $y=2x-1$ and $y=-2x+1$?
(A) The system has one solution
(B) The system has no solution
(C) The system has infinitely many solutions
(D) The system is inconsistent
(E) The system is dependent
- Q5.** The solution to the system $x+y=3$ and $2x-2y=2$ is
(A) $(1, 2)$
(B) $(2, 1)$
(C) $(1, 1)$
(D) $(3, 0)$
(E) $(0, 3)$
- Q6.** What is the x -coordinate of the solution to the system $y=x+2$ and $y=2x-1$?
(A) -1
(B) 1
(C) 2
(D) 3
(E) 0
- Q7.** The system $y=x-2$ and $y=-x+2$ has a solution at
(A) $(0, 0)$
(B) $(1, 1)$
(C) $(2, 0)$
(D) $(0, 2)$
(E) $(1, -1)$
- Q8.** Which of the following systems is equivalent to the system $x+y=2$ and $x-y=-2$?
(A) $y=x+2$ and $y=-x-2$
(B) $y=x-2$ and $y=-x+2$
(C) $x+y=4$ and $x-y=0$
(D) $2x+2y=4$ and $x-y=-2$
(E) $x+2y=2$ and $2x+y=-2$

Open Ended Questions (FRQ)

- Q9.** Solve the system $y=2x+1$ and $y=x-1$. Show all work and explain your answer.
- Q10.** Graph the system $x+y=3$ and $x-y=1$ on the coordinate plane. Label the solution and explain its significance.

Chef's Tips

- Tip 1: Remind students to check their work by plugging the solution back into the system
- Tip 2: Encourage students to use different colors when graphing the lines to avoid confusion
- Tip 3: Suggest that students use the slope-intercept form $y=mx+b$ when graphing the lines